

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0305

Blythe 230 kV
April 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0305. The Interconnection Customer proposes a 617.6 MW natural gas combined cycle facility interconnecting at the Blythe 230 kV substation in the Illinois area. The request was received on February 01, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 201 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,606,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Sweatt-Arch Smith 100.0 kV Ckt 1 is overloaded to 103.8% of its 169 MVA emergency rating for the loss of the Sweatt-Arch Smith 100.0 kV Cir. 2. The project contributes 8.5% DFAX (52.4 MW) to this overload.

The Sweatt-Lawrence Hill 100.0 kV line is overloaded to 103.0% of its 164 MVA emergency rating for the loss of the South Boston-Blythe 230.0 kV L. The project contributes 11.5% DFAX (70.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sweatt-Arch Smith 100.0 kV Line 1	Reconductor line	\$739,000	100% (\$739,000)
Sweatt-Lawrence Hill 100.0 kV line	Reconductor line	\$867,000	100% (\$867,000)
	Total allocated to project		\$1,606,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.